Commercial Profit Intelligence

Customer, Product and Profitability Performance in Supply Chain Operations

Research-style analytical report | October 2026 | Educational case study

Scope and provenance

This report analyzes the user-supplied APL-labelled transaction dataset. The analysis is independent and is not an official APL Logistics or KWE Group financial statement. Currency is unspecified, so all amounts are currency units (CU).

Executive abstract

Across 180,519 transaction lines, 180,519 passed financial validity checks. Valid lines generate 33,054,402.04 CU net revenue and 3,966,902.97 CU recorded profit, giving a revenue-weighted margin of 12.00%. The analysis combines descriptive customer, product and regional contribution with transparent discount sensitivity. It does not infer causality or customer lifetime value.

1. Research questions

RQ1: Where is recorded profit concentrated across customers, categories and markets? RQ2: Which high-revenue segments produce weak margins? RQ3: How do observed discount bands co-vary with profitability? RQ4: What changes under a fixed-volume accounting sensitivity?

2. Data and methodology

Data unit: a transaction line. Financial reconciliation checks Sales minus Order Item Discount against Order Item Total with an absolute tolerance of 0.05 CU. Invalid lines are excluded from financial aggregation and reported separately. Weighted margin is aggregate recorded profit divided by aggregate net revenue. Grouped profit and revenue are additive; group margins are computed from aggregated totals.

Metric	Validated result
Source transaction lines	180,519
Valid financial lines	180,519
Gross sales (CU)	36,784,734.31
Discounts (CU)	3,730,378.40
Net revenue (CU)	33,054,402.04
Recorded profit (CU)	3,966,902.97

Metric	Validated result
Weighted profit margin	12.00%
Loss-making lines	33,784
Loss-making customers	4,069

3. Market contribution

Market	Net revenue (CU)	Profit (CU)	Margin
Europe	9,769,198.23	1,169,442.96	11.97%
LATAM	9,235,761.99	1,123,321.61	12.16%
Pacific Asia	7,434,262.58	857,753.44	11.54%
USCA	4,553,499.87	564,313.78	12.39%
Africa	2,061,679.37	252,071.18	12.23%

4. Product and customer diagnostics

There are 50 categories in the valid sample, with 0 negative-profit categories, and 3 products with negative aggregate recorded profit. 4,069 customer IDs have negative aggregate profit. These are diagnostic flags, not evidence that contracts should automatically be terminated.

5. Discount interpretation and sensitivity

Observed discount-profit relationships may be confounded by product mix, region, customer type and other costs. The what-if slider changes discount dollars by a fixed percentage-point shift of gross sales, bounds discounts between 0 and 100% of gross sales, and holds observed cost, volume and mix constant. This produces accounting sensitivity, not a demand forecast.

6. Recommendations and decision protocol

First reconcile source economics with finance and confirm currency and profit definitions. Next investigate loss-making categories and customers, prioritizing absolute profit exposure and repeatability. Review commercial terms for high-revenue/weak-margin products. Pilot targeted pricing or service changes with a control group where possible. Track realized margin, volume, customer retention and service quality after intervention.

7. Limitations and future research

No order dates, explicit shipping expenses, unique order IDs, acquisition costs, full cost breakdown or controlled pricing experiment are supplied. Therefore temporal trends, shipping-cost allocation, causal discount thresholds, customer lifetime value and price-elasticity forecasts cannot be validated. Future work requires dated transaction and cost data, demand response and a controlled commercial intervention.

Reproducibility

All computations are implemented in analytics.py and exercised by tests/test_analytics.py. The Streamlit application supports global filters, source audit, CSV exports and scenario exploration. See README.md for execution and deployment verification.